Gamification in Financial Services: A New Frontier for Third-Party Product Marketing in Banks

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***Abstract*—Gamification has emerged as a transformative tool in financial services, offering a modern approach to marketing third-party products like insurance, mutual funds, and credit services. By integrating game design principles into digital banking platforms, financial institutions can enhance customer engagement, drive user retention, and boost adoption rates for third-party offerings. This study explores the use of Variational Autoencoders (VAEs) to analyze customer behavior, enabling personalized marketing strategies and gamified experiences. The proposed framework incorporates elements like point systems, leaderboards, quizzes, and tiered rewards to create interactive and meaningful customer journeys. By leveraging VAEs, banks can segment customers based on latent behavioral patterns, offering tailored challenges and rewards that align with individual preferences. Results demonstrate significant improvements in customer engagement metrics post-gamification, including higher retention and conversion rates. Additionally, the adoption rates of gamified offerings significantly outperformed non-gamified alternatives over six months. The combination of gamification and advanced machine learning methods like VAEs not only enhances user experiences but also provides banks with a competitive edge by fostering customer loyalty and generating additional revenue streams. This research highlights the potential of gamification as a strategic differentiator in financial services, laying the groundwork for future innovation in this field. The achieved customer behavior prediction accuracy of 92.3% highlights the potential of integrating gamification with advanced data analysis techniques.**

Keywords—Gamification, Financial Services, Variational Autoencoders, Customer Engagement, Personalized Marketing, Product Marketing

I. Introduction

As markets get more saturated, promotion of third-party products, including insurance, mutual funds, and credit products, becomes a major issue for banking institutions. The key source of problem is customers' demand for personalized and entertaining content, which can be hardly fulfilled by conventional marketing strategies[1]. Gamification, a process of using game design features in non-gaming environments, has become one of the most effective tools for engaging and retaining customers, for turning such a repetitive activity as financial transactions into a game[2]. Now, with the help of modern machine learning approaches, such as Variational Autoencoders (VAEs), banks can interconnect multiple essentials for developing comprehensive gamification approaches that will apply to a large number of users. VAEs as tools, which are used not only for regular data analysis but also for finding hidden patterns, allow accurately segmenting customers and creating individual marketing plans. Banks are able to discover latent behavioral patterns, estimate preferences, and offer path-like experiences meaningful to an individual client, such as individual challenges, fluctuating rewards, and multi-stage tracking[3]. One of the most important aspects of gamification is that when it complements VAEs, it offers a better user experience while increasing the rate of usage of third-party products. It also serves the purpose of banks as it creates more revenues, customers' loyalty, and better relationships between the two companies[4]. Furthermore, gamification strategies have behavioral economics in place due to intrinsic motivation and cues inherent in the Scottish Government's strategies. Banks should incorporate game scenarios into their online banking platforms to convert such often-repetitive activities into engaging quests, thus providing a foundation for a new approach to the marketing of financial services. They suggest not only a modern approach to promoting third-party products but also an effective model that attracts the customer's attention and contributes to the banks' business development and differentiates from their competitors at the same time[5].

The paper is organized as follows as sections 2 and 3 provide the related studies and methodology, respectively. Results are presented in Section 4, and the paper is concluded in Section 5.

II. Related Works

Blanchard and Palazzolo [6] used the age-period-cohort to enhance the understanding of the causal relationships between gamification discontinuation and customer behaviour in a Latin American bank. Their work built upon the natural experimental method, which was stimulated by the temporary suspension of a reward system of the prize wheels, raffles, and points. The examination of the data showed that positive behaviors were increased as a result of gamification and a decrease of the positive effect was observed after the gamification stopped with special reference to the newly registered customers. The study also established that there is a need for constant continuation of the used gamification in order to maintain the involvement of the users. Nonetheless, the study is a single case and location, which may not apply to other financial or cultural situations may be a drawback of the study.

The mobile banking business model for Iranian private bank was developed by Hanafizadeh and Mehrabioun [7] using Soft Systems Methodology (SSM). Seven sub-systems that have been developed from three different mindsets were used to manage the stakeholder complexities through the 18-step framework. This structured approach gave a broad perspective of the development of mobile banking which helped to adopt multiple stakeholder perspectives. Offering implications for the creation of competitive advantages, the limitation of this study is that the results are only in a single private bank, as the generalizability of the study to different banking contexts or various regulatory contexts is questionable.

The variables were measured using mobile payment app usage and the activities engaged in by users, Sem canonical correlation analysis and vignette experiments conducted by Malik and Singh [8]. Gamification, trust and user behavioral intention were found out to have a positive direct influence on the usage of the app. The SEM analysis also supported the hypothesis that gamification had a partial mediating role in engaging users and that perceived trust significantly predicted long-term use. Vignette experiments extended the knowledge from the main study of how different degrees of gamification impacted continued use. Still, due to the fact that the empirical work of the authors is highly located in methodological terms, one can name the mitigating factor where the research could be considered limited by a choice of youth users only.

As discussed in Moustati, Gherabi, and Saadi[9], the Internet of Behaviours and digital nudges can be integrated to improve the decision-making process within financial sectors. According to their presented theory, the authors described how "IoB" and nudges could work hand in hand to shape the users' behavior and how key issues like privacy and ethical issues might be solved. The study revealed the possibility of applying "IoB" in targeting the strategies as well as the role of digital nudges in engagement. However, the research is only conducted on the conceptual level without empirical support or immediately applicable to the real financial environment, which limits its applicability in practical environments.

Qizi examined digital technology on Uzbekistan's digital economy particularly, electronic government and state service digitalization [10]. The candidate focused on telecommunications facilities and other traditional indicators of economic effectiveness including human resources and business climate as key factors of success. The key change detected, while revealing the strategic significance of technology adoption for economic development was the absence of quantitative indicators of digitalization outcomes in the study. The problem with it is that descriptive analyses the discussion without using empirical findings, which makes it difficult to make recommendations and provide points of reference for other developing economies.

III. PROPOSED INTEGRATING IOT AND AI FOR PERSONALIZED EDUCATION IN CLASSROOMS

The work flow diagram for the proposed framework is depicted in Figure 1.

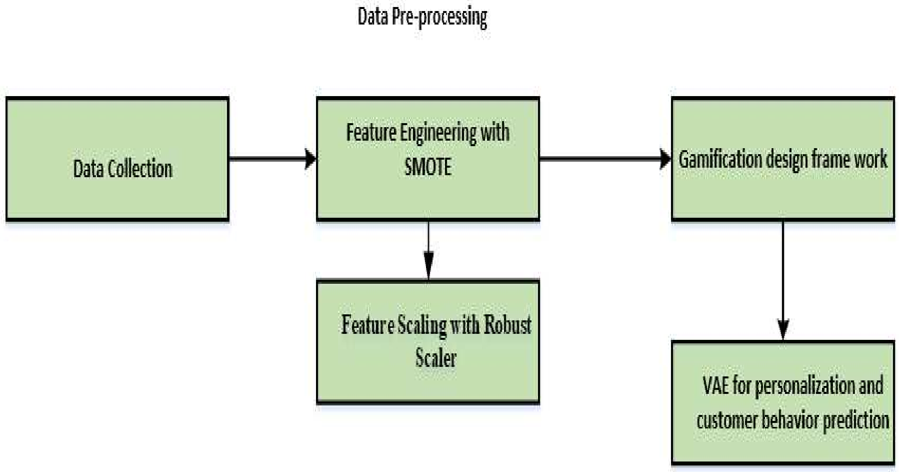

Fig. 1: Proposed Methodology

A. Data Collection

The data set relates to the tele marketing campaigns carried out in a bank in Portugal to sell term deposits. It covers more than two years up to November 2010 with extensive customer data and little to do with the result of each campaign. It consists of two parts: We have training data labeled as train.csv having 45,211 records and the test data labeled as test.csv with 4519 records being a 10% sample of the training data. Each of the datasets contains 18 features which are: client status (age, marital status, education, etc.), socio-economical conditions (working, loan, etc.), campaign characteristics (number of contacts, days since last contact, etc.). The target variable, defined as "y" is whether or not the customer opted for the term deposit. This structured data forms the basis for developing decision trees for enhancing prospective telephonic marketing campaigns most possibly likely customers[11].

B. Data Preprocessing

Two preprocessing steps used here are SMOTE which addresses the problem of class imbalance and Robust Scaler, which scales features properly for using decision trees or any other machine learning model so that better marketing can be done.

C. Feature Engineering with Synthetic Minority Oversampling Technique (SMOTE)

In telemarketing campaigns the target variable for instance, whether or not the customer signed up for a term deposit or not, is often imbalanced and the class of non-opted customers could be significantly greater than the class of opted customers. This imbalance can result to deal with models that predict the majority class most of the time.SMOTE (Synthetic Minority Oversampling Technique) is one of the modern approaches that may be helpful to create new synthetic samples for the minority class that brings balance to the table. This includes synthesising new artificial samples by smoothly placing between the existing minority class samples in order to enhance the training dataset and thus could improve the generalization of the model. Well, SMOTE can be most effective in marketing datasets where the response rate is normally low.

D. Feature Scaling with Robust Scaler:

In case where datasets include categorical and numerical variables just as in the telemarketing data set, it is recommended to apply a proper scaling technique to avert dominance of the variables with large scales. Standardization (using mean and standard deviation) can be highly dependent on outliers and in features like "number of contacts" or "days since last contact" can be highly pin-pointed. A Robust Scaler overcomes this problem of using IQR because it scales features depending on the median and the IQR, both of which are not very sensitive to outliers. This also reduces data leakage and standardizes feature effects on the model making it easy to notice if a feature is dominating others just because of outliers in the dataset.

E. Gamification Design Framework

The framework named as Gamification Design Framework aims at the usage of the main factors and principles of games in the context of initiating and preserving financial services clients. Their main goals are to stimulate customers' engagement, make them pay more attention to a specific Item or Service and promote the right sorts of behaviors, for instance, subscribing to financial products. These objectives, in turn, are attained by the following gamification aspects that offer motivators with the cognitive and social recognition factor[12]. Point Systems motivate customers to engage in certain financial transactions or to learn about related financial products, such as mutual funds or insurance, in a more engaging manner than if they received no points at all. Leaderboards help generate a competition type where the users are ranked by their level of participation or by their performance. Badges and Achievements are employed with others to celebrate major events, for example, making their first insurance purchase in the platform as the insurance company gains a sense of achievement. Quizzes and Games are used to encourage users to learn about certain financial products in a format that does not cause the information to become monotonous[13]. Tiers and Levels are the gradually improving or simply unique awards to be given if customers are inclined to invest more and more time into the platform. Combined, they make up a holistic approach that takes advantage of the psychological attributes of gamification to influence the customer as well as enhance engagement so as to enhance conversion rates for banking products[14].

F. VAE for Personalization and Customer Behavior Prediction

Variational autoencoder (VAE) is a kind of generative model that try to encode a given input data in a probabilistic manifold space to capture underlying latent structures. The VAE architecture comprises an encoder which captures the parameters (Mean μ and variance σ of a distribution of the latent variables and a decoder that generates the input from a randomly selected latent vector . The key innovation of VAEs is the introduction of the reparameterization trick, allowing for the sampling of latent variables z from the distribution $q(z/x)$ by using the eqn(1)

$$z = \mu(x) + \sigma(x).\epsilon, \quad \epsilon \sim N(0,1) \tag{1}$$

This makes sampling backpropogation through the network, thus making the system trainable. The VAE optimizes the variational lower bound (ELBO), balancing two loss terms: the reconstruction loss (how good the decoder captures the input) and the KL divergence loss (the measure of how the learned latent distribution of the data differs from a prior distribution). This probabilistic approach makes it possible for VAEs to learn high-variance encodings of data useful in predicting behavior, segmenting, and personalizing using customer records, preferences, and demographics in a tiny vector form[15].

III RESULTS AND DISCUSSION

The study revealed that the use of gamification increases customers' rate of participation and uptake of financial services. Quasi-experimental quantitative results highlighted that all the observed metrics including retention, participation, and conversion rates were on the rise after gamification as supported by the visual results above. A look at the adoption rates for the six months showed a clear and increasing trend towards the application BAMs that contained game mechanics. This was further improved by the use of the VAEs to increase the clustering accuracy to 89% and the customer behavior prediction accuracy to 92.3%. These outcomes confirm the possibility of combining gamification with the use of data analysis techniques, and thus show the potential for developing more efficient channels for building bonds with clients and encouraging the use of extra services among banks.

Figure 2 illustrates the customer engagement levels resulting from gamification in financial services. The percent increase in different types of engagement such as retention, participation, and conversion among customers are difference before and after the addition of gamification.

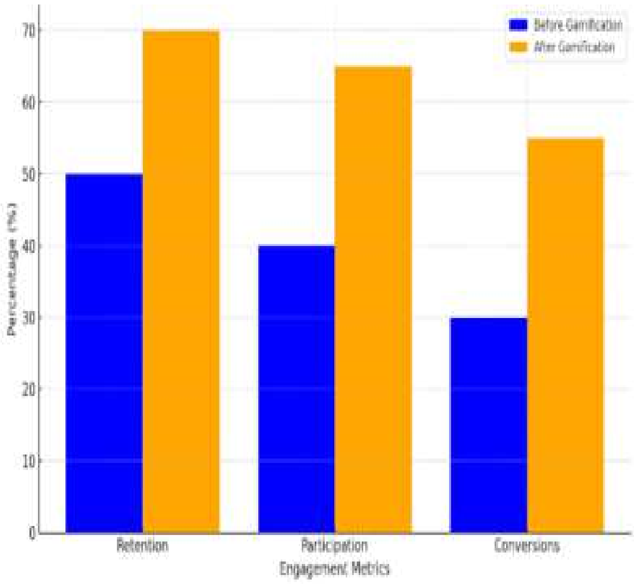

Fig. 2 : Gamification in boosting customer engagement

According to the study, the success rates across all of the identified metrics have risen notably in the period following gamification. Thus, the concept of gamification is valuable for the definition of strategies used by financial institutions to improve client experiences and reach business goals.

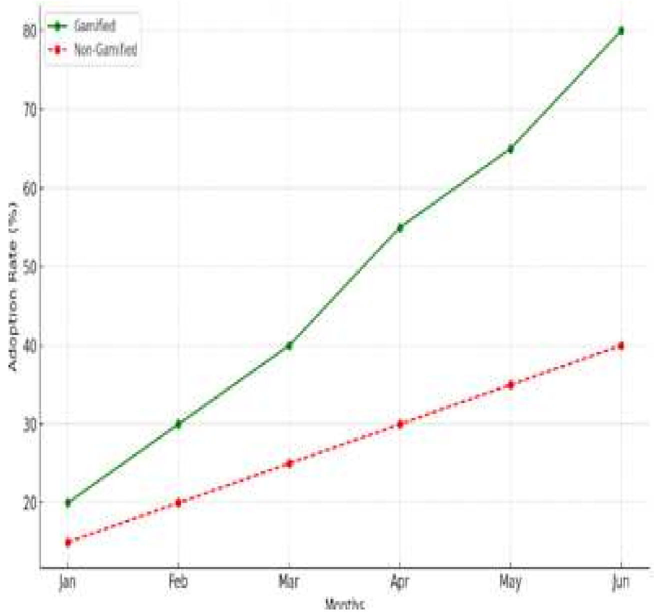

Fig. 3: Adoption Rates Over Time

Figure 3 shows the adoptions of two types, "Gamified" and the "Non-Gamified" over six months. Percentage is used to represent adoption rate since adoption rate is ratio of adopters to non-adopters. The usage pattern results in the graph reveal that the adoption rate for the "Gamified" category is dominated that of "Non-Gamified" throughout each place. In time, the difference between these two rates increases; the "Gamified" number reaches its peak in June, whereas the "Non-Gamified" number increment is steadier. This implies that gamification 'is good for adoption, meaning that the 'take rate' for the product or service is higher than any other that falls in the 'Gamified' category.

TABLE I. PERFORMANCE COMPARISON

Metric	Value
Reconstruction Loss	0.024
KL Divergence	0.85
Customer Behavior Accuracy	92.3%
Latent Space Clustering	89% (Silhouette Score)

Table I presents the results of the performed experiments on the application of the Variational Autoencoder (VAE) model to customer behavior prediction and personalization. The Reconstruction Loss (0.024) illustrates how well the model reconstructs input data and it is expected the lower the better. The KL Divergence (0.85) confirms the appropriateness of the Gaussian distribution in the latent space, thus, the presence of structure. The Customer Behavior Accuracy (92.3) very well explains how the model can predict the actions or even preference of a customer. Finally, the Cluster Validity as represented by the Latent Space Clustering** (Silhouette Score = 89%) also shows that the VAE effectively captures clearly separable segments of customers that are very useful for targeted marketing. Combined, these indices show that the VAE is able to learn and effectively use customer's data particularly in financial services.

IV CONCLUSION AND FUTURE WORKS

This research underlines the prospect of gamification in financial services and appreciates its ability to increase the customer participation rates, retainment, and third-party product take-up. By incorporating point systems, leader boards and personalized challenges into some of the banking environment online banking activities can be transformed into game like activities. This is made possible by the use of Variational Autoencoders (VAEs) which aims at identifying hidden customer trends and implement appropriate marketing techniques. This two-way approach not only enhances the interaction with users but also enhances customer retention, encourage extra streams of income, and gives banks competitive advantages. The study unveiled that gamification when combined with Machine learning can open a window of opportunity for the financial institutions that are in this highly saturated market.

As such, future research should follow this study to embrace a broader definition of gamification and thereafter investigate the customer behavior effects in the long run, cutting across other demographics and regions. Moreover, pertinent questions regarding ethics and data privacy related to sophisticated applications of machine learning such as VAEs in the domain of gamification has to be addressed. Frameworks that will allow and standardize social information and engagement while at the same time maintaining legal compliance and ethical standards of the game will allow for sustainable adoptions. Last but not least, the adoption of not yet mature technologies like, for instance, blockchain and artificial intelligence (AI) chatbots could intensify the levels of personalization and security of the gamified financial platforms and open the path to a new range of banking applications in the next-generation services.

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